

Protocol Communication

What are the different communication protocols?

Which ones were built for the internet?

What are each protocols limitations?

How much overhead do they have?

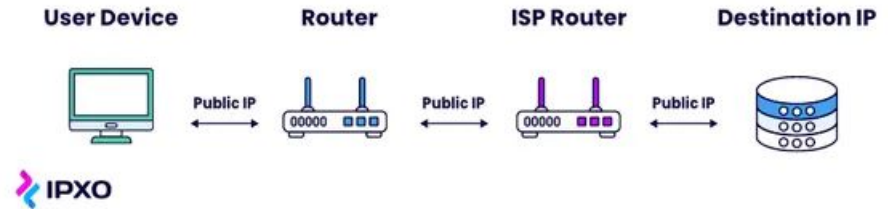
How did they reach consensus?

What problems do they solve?

What type of radio waves?

How similar are they?

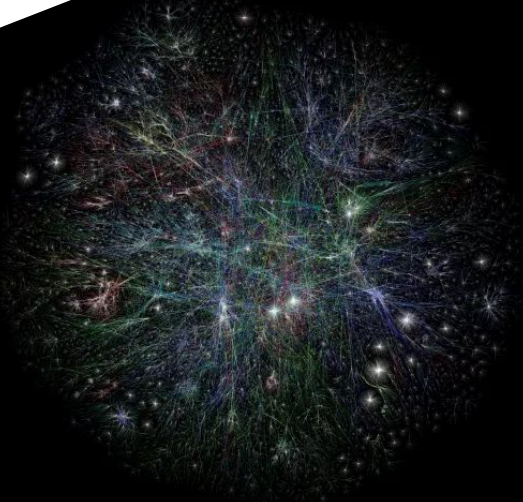
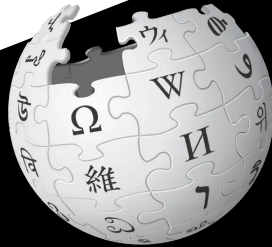
Why were they added?



The Internet

The internet is the biggest and most important communication system ever created. Modern communication technology can only be understood in reference to it.

"The internet is the global system of interconnected computer networks that uses the Internet protocol suite (TCP/IP) to communicate between networks and devices."



Internet Protocol Suite Layers

Communicating 1s and 0s from any machine to any other machine using radio waves and wire is very complicated

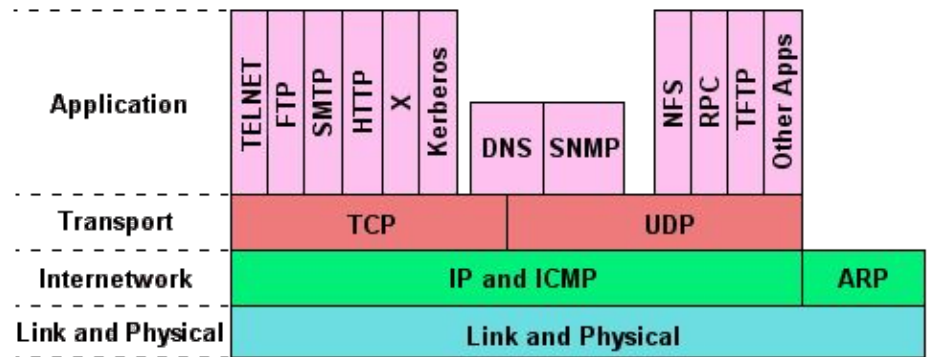
Because of this, multiple layers are used:

Link - device to device

Internet - proxy devices

Transport - reliability

Application - everything else



Link Layer

"The link layer is the group of methods and communications protocols confined to the link that a host is physically connected to."



The link layer includes many different technologies, not just one. If all physical connections on the internet were wires, the link layer would only be one protocol. Since this is not the case, the link layer refers to all protocols for all technologies.

Internet Layer

It is impractical to connect every device on a large network
To connect all the devices, data needs to be transported via
gateways

The internet layer is a group of internetworking methods,
protocols, and specifications that does this

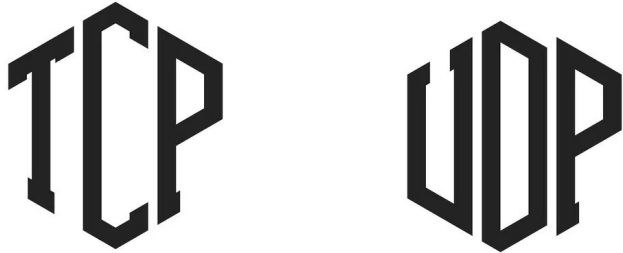


IP ADDRESS

Transport Layer

Reliably sending data is core to the functionality of the internet. Every website could have its own way of ensuring reliability.

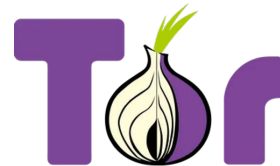
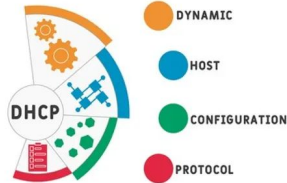
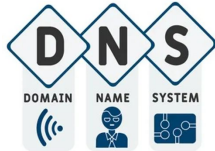
Since reliability is core to the internet, it is standardized in the transport layer.



Application Layer

The application layer is all the functionality two applications need to communicate that hasn't been handled in either the link, internet, or transport layer

The scope of communication functionality is broad



Internet Development Notes

1753 Theoretical electrostatic telegraph with wires for each letter

1774 Short distance electric telegraph realized

1800 Voltaic Pile

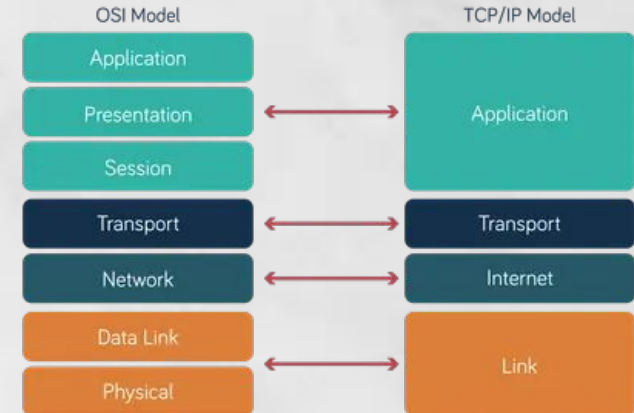
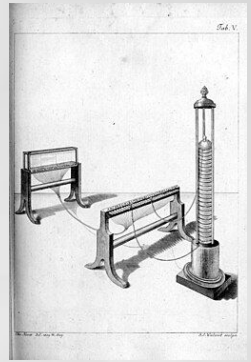
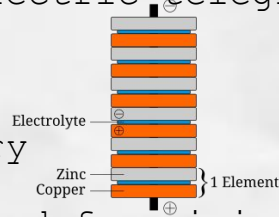
1948 Information Theory

1969 ARPANET was created for giving access to computation

1978 TCP and IP separated

1982 Origin of the term Internet

1984 OSI model published



Bluetooth

Bluetooth does not use the internet protocol suite

It is a short distance wireless communication standard for directly connecting devices

Bluetooth uses packets

